Apurva Patkeshwar

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Software Engineer with 10+ years of experience building distributed systems, ML infrastructure, and large-scale platforms. Led initiatives across real-time decisioning, model serving, evaluation, reliability, and rollout strategy.

Experience

Uber — New York, NY
Software Engineer — Feb. 2023 – Present

- Led multi-year access control transformation from static policies to adaptive ML driven authorization, across risk visibility, AI assisted human-in-the-loop decisions, and ML-based enforcement for agentic AI workflows. Established AI Security team's ML roadmap and led strategy and experimentation
- Architected adaptive authorization system at AI gateway with real-time ML inference and challenge-based enforcement. Built SHAP-guided explanations turning model signals into user-facing rationale. Enabled risk-based decisions auto-approving 85% of low-risk requests, escalating high-risk interactions
- Owned and launched Employee Security Score globally across 85K employees, aligned Legal, Security, Privacy, and Risk stakeholders. Improved phishing resilience 6.3x, reduced high-risk attachment opens 71%, shipped a GenAI support chatbot adopted by 23K employees with 4.45/5 CSAT [Python, Hive]
- Defined ML-based access approval strategy by introducing cohort-based Smart Approvals, reducing manual review toil by 60%. Automated 6-hour manual ML deployment with CI/CD for faster iteration
- Led privileged access remediation across 25+ service owners. Eliminated contractor superuser access company wide, reduced FTE standing access by 91%, uncovered 1.7K hidden elevated access paths
- Established ML safety infrastructure gating model promotion by sanitizing 37K historical labels, benchmarking 13K edge cases using LLM, defining evaluation standards for partial AUC, F0.5, subgroup fairness, data drift
- Migrated graph embedding pipeline by precomputing walk weights and FAISS neighbor search; from 19-hour runtime to 30 mins on distributed Spark with seeded reproducibility, enabling production scale iteration

Twitter — Seattle, WA
Machine Learning Engineer — Jul. 2021 – Jan. 2023

- Delivered a fully managed, config-driven ML serving platform for model deployment, rollout, and inference at scale with 20 ms p99 latency [Scala, Python, Hadoop, Kubernetes, Triton, YAML]
- Developed a model observability system with cross-functional collaboration for drift monitoring and alerting, surfacing regressions and reducing average retraining frequency by 60% while preserving model quality

Amazon — Seattle, WA
Software Development Engineer — Jun. 2018 – Jul. 2021

- Decoupled Alexa's audio streaming path from core ASR architecture, cached audio closer to customers moving speech processing closer to the edge. Reduced p99 user perceived latency by 25% in emerging markets [AWS]
- Led migration from push-to-talk to wake-word models by turning off endpointing in cloud. Rerouted 1M daily requests, reduced operational load, shortened rebuild execution by 9 hours. Saved $500K/year in model release cost, $170K/year in model rebuild cost, and $120K/year in benchmarking costs [Java, XML]

Microsoft — Hyderabad, India
Software Engineer — Jul. 2014 – Jul. 2016

- Built an office-space web app visualizing real-time occupancy, noise, temperature data using custom 3D-printed sensors. Provided RFID-based room reservation, driving a 3x increase in monthly usage [C#, SQL, .NET]

Education

New York University - Courant Institute of Mathematical Sciences — New York, NY
Master of Science in Computer Science, 3.7/4.0 — May 2018

University of Mumbai - Sardar Patel Institute of Technology — Mumbai, India
Bachelor of Engineering in Information Technology, 3.6/4.0 — Jun. 2014

Awards

Uber, Security Eng Leadership Award | **Google**, Code Jam Top 150 worldwide | **Forbes**, Under 30 Scholar

Leadership

Uber, Immigrants ERG Mentor and US&C Lead | **McKinsey**, Business Leadership Program
Twitter, Employee Ambassador | **Microsoft**, MACH Education Committee Lead